

Micah Gao

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EDUCATION

Indiana University | Luddy School of Informatics

Bloomington, IN

B.S. in Informatics; Kelley Accounting Certificate & Business Minor

Expected May 2028

EXPERIENCE

Software Engineering & Corporate Accounting Intern

May 2026 – Present

H.T. Hackney | \$4B+ Wholesale Distributor

Indianapolis, IN

- Uncovered a systemic formula error behind \$30K in recurring overpayments while processing tariff refunds across 1,600+ containers and \$4M+ in order value, then delivered a corrected formula adopted for all future refund cycles
- Built an Excel refund model and pivot analysis from raw invoice and container data that executives up to the company president use to set item-level refund amounts and anchor negotiations with the company's largest accounts
- Valued \$500K+ in prepaid overseas inventory for month-end close by matching 300+ vendor invoices in Microsoft Dynamics to container codes, warehouses, and landed costs, then preparing accruals across 20+ overseas vendors
- Develop a self-hosted internal e-commerce catalog in Next.js that lets employees buy furniture at wholesale prices

Machine Learning Research Assistant

April 2026 – Present

University of Indianapolis | MBA Artificial Intelligence Laboratory

Indianapolis, IN

- Lead 1 of 2 MBA-level M&A research tracks, computing 15 risk and solvency ratios from SEC filings of 50+ acquisitions
- Apply entropy-based information-gain feature selection across 30 variables to rank the 5 strongest deal predictors
- Research findings are integrated into the machine learning curriculum taught across the university's MBA program

Director of Technology

June 2026 – Present

Aperture Partners

Bloomington, IN

- Lead technical workshops training consultants and project leads for a 501(c)(3) B2B consulting organization named Kelley Organization of the Year, serving clients including Toyota, Cummins, Bond AI, and the Indianapolis Indians
- Rebuilt the organization's Slack workspace, channel architecture, and onboarding flow for a fall class of up to 40 new consultants, and designed its tiered client pricing framework spanning \$2K to \$10K+ per client engagement

PROJECTS

StudForge | Swarm Robotics World Compiler; Python, React Three Fiber, Qwen, Redis

June 2026

- Built the real-time 3D frontend in TypeScript and React Three Fiber, texturing compiler-generated brick worlds with Midjourney MCP assets and rendering live swarm simulation so users can inspect each validated scene as it compiles
- Engineered the Python compiler backend on typed Pydantic contracts, turning prompts into a schema-constrained SceneSpec and gating every layout through connectivity, collision, and coverage validation before canonical USD export
- Approximated an NP-hard set-cover compression with a fine-tuned Qwen2.5-1.5B model held to a deterministic validator floor, and ran three observer agents streaming findings into Redis-backed memory that biased later generations

Onkos | Agentic Oncology Trial Matcher; Next.js, FastAPI, LangGraph, vLLM

April 2026

- Engineered an agentic clinical trial matching system pairing cancer patients to active oncology trials, scoring each against 20+ eligibility criteria and surfacing per-criterion disqualifying logic with cited rationale for clinician review
- Self-hosted MediX-R1, a 30B medical vision-language model, on an NVIDIA GH200 via vLLM, and orchestrated Kimi-K2 through LangGraph for agentic tool calls and RAG-grounded clinical reasoning over a corpus of trial protocols

HONORS & PROGRAMS

Grand Prize Finalist | UC Berkeley AI Hackathon

June 2026

UC Berkeley

Berkeley, CA

- Named a top 10 finalist out of 460+ teams and 1,600+ participants at the world's largest AI hackathon, run by Cal Hacks
- Pitched StudForge live on the main stage to venture capital judges and founders, placing 2nd in the Engineering Track

Regeneron Grand Prize Winner | HackPrinceton

April 2026

Princeton University

Princeton, NJ

- Won the \$1,000 Grand Prize on Regeneron's Best Use of AI for Clinical Trials track against 600+ competing hackers

Anthropic Runner-Up | Stanford TreeHacks

February 2026

Stanford University

Palo Alto, CA

- Selected from 15,000+ applicants at a 3% acceptance rate, one of the most competitive collegiate hackathons
- Demoeed Redhue, a wildfire-spread dashboard, to Anthropic and a CAL FIRE chief as 1 of 20 projects picked from 400+

Selected Participant | Deloitte Compass Conference

August 2026

Deloitte

Westlake, TX

- Selected nationally through two rounds of interviews for Deloitte's fully funded, three-day early-career conference at Deloitte University, the firm's leadership center, for casework and mentorship alongside Deloitte partners and advisors

External Programs: PEAK6 Young Professionals, Expedition EY Technology Track, Goldman Sachs Possibilities Series

TECHNICAL SKILLS

Business: Excel, PowerPoint, Microsoft Dynamics, financial modeling, SEC 10-K/10-Q analysis, M&A research

Engineering: Python, TypeScript, JavaScript, Swift, SQL, React, Next.js, FastAPI, AWS, Supabase, Postgres, Docker

ML/AI: RAG, vision-language models, agentic systems, LLM fine-tuning, feature engineering, vLLM, LangGraph